SESSION PLAN

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DATE:	23 Oct 2026
PROGRAMME TITLE	Advanced Machine Learning in Production: From Leakage-Free Pipelines to Deployed Models
DURATION	4 Hours
LEARNING OUTCOMES	By the end of the session, the participants will be able to: a) Select the appropriate approach (classic machine learning, deep learning or a large language model) for a given business problem b) Detect data leakage and build a leakage-free scikit-learn pipeline with a validation strategy that fits the data c) Organise training code into modular, configurable and reproducible scripts d) Tune hyperparameters with Optuna and track experiments with MLflow e) Explain model predictions with SHAP and translate them into business language f) Detect data drift using the Population Stability Index (PSI) and the Kolmogorov-Smirnov (KS) test, and set alert thresholds g) Serve a trained model as a REST API with FastAPI inside a Docker container h) Describe what to monitor in production and how the same workflow runs on Azure Machine Learning

Modules / Learning Points	Resources/ Training Aids	Method(s)/ Activities	Duration (minutes)
Introduction and Ice Breaking • Welcome, trainer and participant introductions (4 min) • Programme overview, objectives and ground rules (3 min) • Ice breaking: “Would you ship it?” A model scores 99% accuracy on a dataset where 1% of loans default (5 min) • Environment check, so every participant can run the first notebook cell (3 min)	• Welcome slides • Programme outline • One-slide ice-breaker scenario • Setup guide in the course repository README	• Facilitated introductions • Ice breaking activity (poll and short discussion) • Checkpoint: every participant runs the first notebook cell without error	15

Modules / Learning Points	Resources/ Training Aids	Method(s)/ Activities	Duration (minutes)
Module 1: Trustworthy Data and Leakage-Free Pipelines • Choosing the right approach: where classic machine learning, deep learning and large language models fit, mapped to real business problems (10 min) • Data quality and data leakage: the main data quality dimensions, and target, train-test, temporal and group leakage (10 min) • Finding leakage: hands-on search for every leakage source in a loan-default dataset (15 min) • Leakage-free pipelines and validation: the scikit-learn Pipeline, splitting before preprocessing, and time-ordered validation (20 min) Outcome focus: learning outcomes (a) and (b)	• Lecture notes / slides (Sessions 01 and 02) • Session 02 notebook (data quality and leakage) • Loan dataset (sample_loans.csv) • scikit-learn	• Mini-lecture • Group discussion • Live demonstration • Hands-on exercise • Checkpoint: each participant's pipeline is scored on a time-ordered split with no leaked feature	55
Module 2: Production Training, Tuning and Tracking • From notebook to modular scripts: single-responsibility modules, YAML configuration, seeds and logging, and running the training entry point from the command line (15 min) • Hyperparameter optimisation: grid, random and Bayesian search (Bergstra & Bengio, 2012), with hands-on tuning in Optuna (20 min) • Experiment tracking: logging parameters, metrics and artefacts with MLflow, and comparing runs (20 min) Outcome focus: learning outcomes (c) and (d)	• Lecture notes / slides (Sessions 03 and 04) • Session 03 training scripts (main.py and config.yaml) • Session 04 notebook (hyperparameter optimisation) • Optuna and the MLflow tracking UI	• Mini-lecture • Code walkthrough • Hands-on exercise • Checkpoint: main.py trains from the command line and the best Optuna trial appears in the MLflow UI	55

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Module 3: Explainability and Drift Monitoring • Explaining models with SHAP: global summary plots and local waterfall plots (Lundberg & Lee, 2017) (20 min) • From explanation to business narrative: writing a plain-language explanation of one prediction for a non-technical stakeholder (10 min) • Detecting data drift: feature, concept and label drift, measured with PSI and the KS test (15 min) • Anomaly detection and alert thresholds: trainer demonstration of Isolation Forest, then setting thresholds and a monitoring cadence (10 min) Outcome focus: learning outcomes (e) and (f)	• Lecture notes / slides (Sessions 05 and 06) • Session 05 notebook (explainability) • Session 06 notebook (drift and anomaly detection) • SHAP library	• Mini-lecture • Live demonstration • Hands-on exercise • Pair review • Checkpoint: one written plain-language explanation of a single prediction, and a PSI table with drifted features flagged	55
Module 4: Serving, Monitoring and Cloud • Serving a model as a containerised API: FastAPI with Pydantic validation, built and run in Docker and tested with a curl request (20 min) • The MLOps dashboard: model performance, data health, system health and business impact, read from a six-panel dashboard (15 min) • Moving to Azure Machine Learning: trainer demonstration of the workspace, a cloud training job and the model registry (15 min) • Back-to-work action plan and wrap-up: personal next steps, questions, key takeaways and feedback (10 min) Outcome focus: learning outcomes (g) and (h)	• Lecture notes / slides (Sessions 07 to 10) • Session 07 API application (FastAPI app and Dockerfile) • Docker Desktop • Session 08 notebook (MLOps dashboard) • Azure Machine Learning studio (trainer's workspace) • Personal action plan template • Programme feedback form	• Live demonstration • Hands-on exercise • Group discussion • Individual action planning • Checkpoint: the container returns a prediction to a curl request, and each participant leaves with a personal action plan	60
Total number of minutes: 240			